

So, I made this lab with DevOps.Noderino App

All docker images need:

```
joao@jocapc:~/devops.noderino/prometheus$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
e8bdc5f34190	prom/prometheus:latest	"/bin/prometheus --c..."	4 minutes ago	Up 4 minutes	0.0.0.0:9090->9090/tcp, [::]:9090->9090/tcp	dev
opsnoderino_prometheus_1	curlimages/curl:latest	"/entrypoint.sh sh -..."	8 minutes ago	Up 8 minutes		dev
opsnoderino_loadgen_1	nginx/nginx-prometheus-exporter:latest	"/usr/bin/nginx-prom..."	8 minutes ago	Up 8 minutes	0.0.0.0:9113->9113/tcp, [::]:9113->9113/tcp	dev
opsnoderino_nginx-exporter_1	prom/alertmanager:latest	"/bin/alertmanager -..."	8 minutes ago	Up 8 minutes	0.0.0.0:9093->9093/tcp, [::]:9093->9093/tcp	dev
a2d78829c624	gcr.io/cadvisor/cadvisor:latest	"/usr/bin/cadvisor -..."	8 minutes ago	Up 8 minutes (healthy)	0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp	dev
opsnoderino_cadvisor_1	devopsnoderino_slow-api	"docker-entrypoint.s..."	8 minutes ago	Up 8 minutes	8080/tcp	dev
d80cd36c3301	devopsnoderino_slow-api_1	"docker-entrypoint.s..."	8 minutes ago	Up 8 minutes	8080/tcp	dev
c0fbfae1307f	devopsnoderino_slow-api_3	"docker-entrypoint.s..."	8 minutes ago	Up 8 minutes	8080/tcp	dev
opsnoderino_slow-api_3_1	devopsnoderino_slow-api_2	"docker-entrypoint.s..."	8 minutes ago	Up 8 minutes	8080/tcp	dev
684aade5768	devopsnoderino_slow-api_2_1					

Prometheus UI:

nodejs_eventloop_lag_mean_seconds	instance	job
0.010175517257142856	slow-api_3-8080	slow-api
0.010173076375310204	slow-api_2-8080	slow-api
0.010170602057142857	slow-api_8080	slow-api

All the targets are up:

Service	Endpoint	Labels	Last scrape	State
cadvisor	http://cadvisor:8080/metrics	instance="cadvisor:8080" job="cadvisor"	1.773s ago 7ms	UP
nginx	http://nginx-exporter:9113/metrics	instance="nginx-exporter:9113" job="nginx"	458ms ago 15ms	UP
slow-api	http://slow-api_3-8080/metrics	instance="slow-api_3-8080" job="slow-api"	2.643s ago 1ms	UP
	http://slow-api_8080/metrics	instance="slow-api_8080" job="slow-api"	2.156s ago 1ms	UP
	http://slow-api_2-8080/metrics	instance="slow-api_2-8080" job="slow-api"	2.208s ago 1ms	UP

Docker-compose:

```
version: "3.8"
```

```
networks:
```

```
  monitoring:
```

```
    driver: bridge
```

```
services:
```

```
  slow-api:
```

```
    build: ./app
```

```
    environment:
```

```
      - PORT=8080
```

```
      - DESIRED_PATH=/work
```

```
      - NUMBER=1
```

```
  deploy:
```

```
    resources:
```

```
      limits:
```

```
        memory: 80m
networks:
  - monitoring

slow-api_2:
  build: ./app
  environment:
    - PORT=8080
    - DESIRED_PATH=/work
    - NUMBER=2
  deploy:
    resources:
      limits:
        memory: 80m
    networks:
      - monitoring

slow-api_3:
  build: ./app
  environment:
    - PORT=8080
    - DESIRED_PATH=/work
    - NUMBER=3
  deploy:
    resources:
      limits:
        memory: 80m
    networks:
      - monitoring

nginx-lb:
  image: nginx:alpine
  volumes:
    - ./nginx/nginx.conf:/etc/nginx/nginx.conf:ro
  ports:
    - "8000:80"
  networks:
    - monitoring

nginx-exporter:
  image: nginx/nginx-prometheus-exporter:latest
  command:
    - -nginx.scrape-uri=http://nginx-lb/nginx_status
```

```
ports:
  - "9113:9113"
depends_on:
  - nginx-lb
networks:
  - monitoring

cadvisor:
  image: gcr.io/cadvisor/cadvisor:latest
  privileged: true
  volumes:
    - /:/rootfs:ro
    - /var/run:/var/run:ro
    - /sys:/sys:ro
    - /var/lib/docker:/var/lib/docker:ro
  ports:
    - "8080:8080"
  networks:
    - monitoring

prometheus:
  image: prom/prometheus:latest
  volumes:
    - ./prometheus/prometheus.yml:/etc/prometheus/prometheus.yml:ro
    - ./prometheus/alerts.yml:/etc/prometheus/alerts.yml:ro
  ports:
    - "9090:9090"
  depends_on:
    - nginx-exporter
    - cadvisor
  networks:
    - monitoring

alertmanager:
  image: prom/alertmanager:latest
  volumes:
    -
./alertmanager/alertmanager.yml:/etc/alertmanager/alertmanager.yml:ro
  ports:
    - "9093:9093"
  networks:
    - monitoring
```

```
loadgen:
  image: curlimages/curl:latest
  command: >
    sh -c "while true; do
      curl -s http://nginx-lb/work > /dev/null &
      sleep 0.05;
    done"
  depends_on:
    - nginx-lb
  networks:
    - monitoring
```